

AWS VPC – Create VPC, Subnets, Route Table, EC2 Instances and Connectivity Test

Task:

Create a VPC, Internet Gateway, public and private subnets, route table, launch two EC2 instances, and test connectivity between them.

(a) Create VPC:

1. Login to AWS Management Console
Open <https://aws.amazon.com/> and sign in
2. Open **VPC Dashboard**
Search and select VPC
3. Click **Create VPC**
4. Configure:
 - Name: MyVPC
 - IPv4 CIDR block: 10.0.0.0/16
 - Leave other settings default
5. Click **Create VPC**

(b) Create Internet Gateway:

1. Go to **Internet Gateways**
2. Click **Create internet gateway**
3. Name: MyIGW
4. Click **Create**
5. Attach to VPC:
 - Select IGW

- Click **Actions** → **Attach to VPC**
- Select `MyVPC`
- Click **Attach**

(c) Create Subnets:

Public Subnet:

1. Go to Subnets → Create subnet
2. VPC: `MyVPC`
3. Name: `PublicSubnet`
4. CIDR: `10.0.1.0/24`
5. Click **Create**

Private Subnet:

1. Click **Create subnet** again
2. Name: `PrivateSubnet`
3. CIDR: `10.0.2.0/24`
4. Click **Create**

(d) Create Route Table:

1. Go to Route Tables
2. Click **Create route table**
3. Name: `PublicRouteTable`
4. VPC: `MyVPC`
5. Click **Create**
6. Edit routes:
 - Add route: `0.0.0.0/0` → Internet Gateway (`MyIGW`)
7. Associate subnet:
 - Attach `PublicSubnet`

(e) Create EC2 Instances:

Instance 1 (Public Subnet):

1. Go to EC2 → Launch Instance
2. Name: PublicInstance
3. Subnet: PublicSubnet
4. Auto-assign Public IP: Enabled
5. Launch instance

Instance 2 (Private Subnet):

1. Launch another instance
2. Name: PrivateInstance
3. Subnet: PrivateSubnet
4. Auto-assign Public IP: Disabled
5. Launch instance

(f) Connect to Public Instance:

```
ssh -i mykey.pem ec2-user@<Public-IP>
```

(g) Try Ping Between Instances:

Get private IP of PrivateInstance

Example: 10.0.2.10

From PublicInstance, run:

```
ping 10.0.2.10
```